Improving the Performance of a Poisson Solver

Matouš Zatloukal

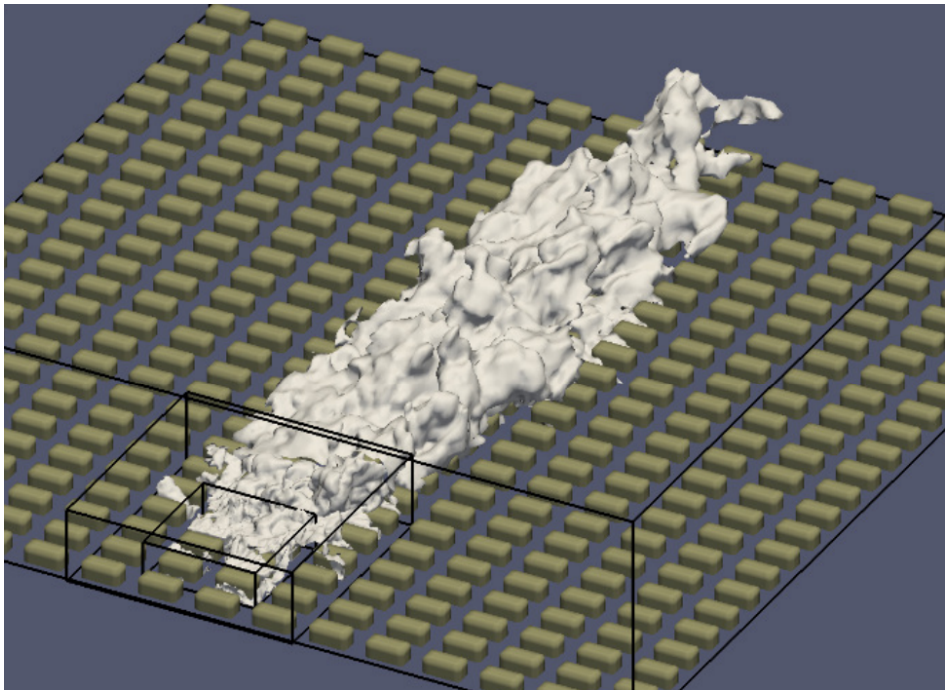
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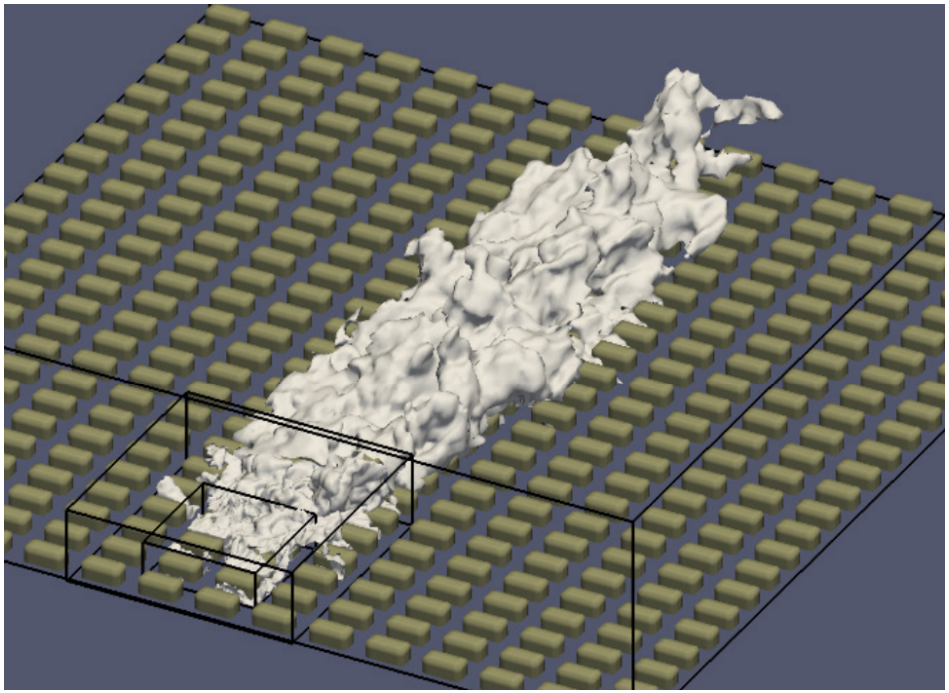
ELMM

- Extended Large Eddy Microscale Model (ELMM) is a model for atmospheric flows
- Written in Fortran
- Parallelized using OpenMP, MPI, or hybrid MPI + OpenMP



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- In each time step a Poisson equation must be solved

PoisFFT - Poisson solver

- Solves the Poisson equation using FFT-based methods
- Parallelized the same way as ELMM
- Best starting point for ELMM performance improvement

Outside view of PoisFFT

- User defines a domain:

Outside view of PoisFFT

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  - Dimensionality
  - Number of grid cells
  - Grid lengths
  - Boundary conditions

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- Then (repeatedly) solves the equation

Inside view of PoisFFT

PoisFFT has two phases:

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1. Initialization

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1. Initialization
   - Precompute the normalization factor and eigenvalues
   - Prepare FFT plan (or plans)
   - Allocate workspaces
2. (repeated) Execution
   - Apply forward FFT
   - Divide every element by a sum of precomputed eigenvalues
   - Apply backward FFT
   - Normalize the result

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  - Poor architecture design
  - Sparsely commented code
  - Bad code hygiene

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poisson-solver - a modern PoisFFT port

Our goal

Create a PoisFFT GPU port with better performance and better software architecture

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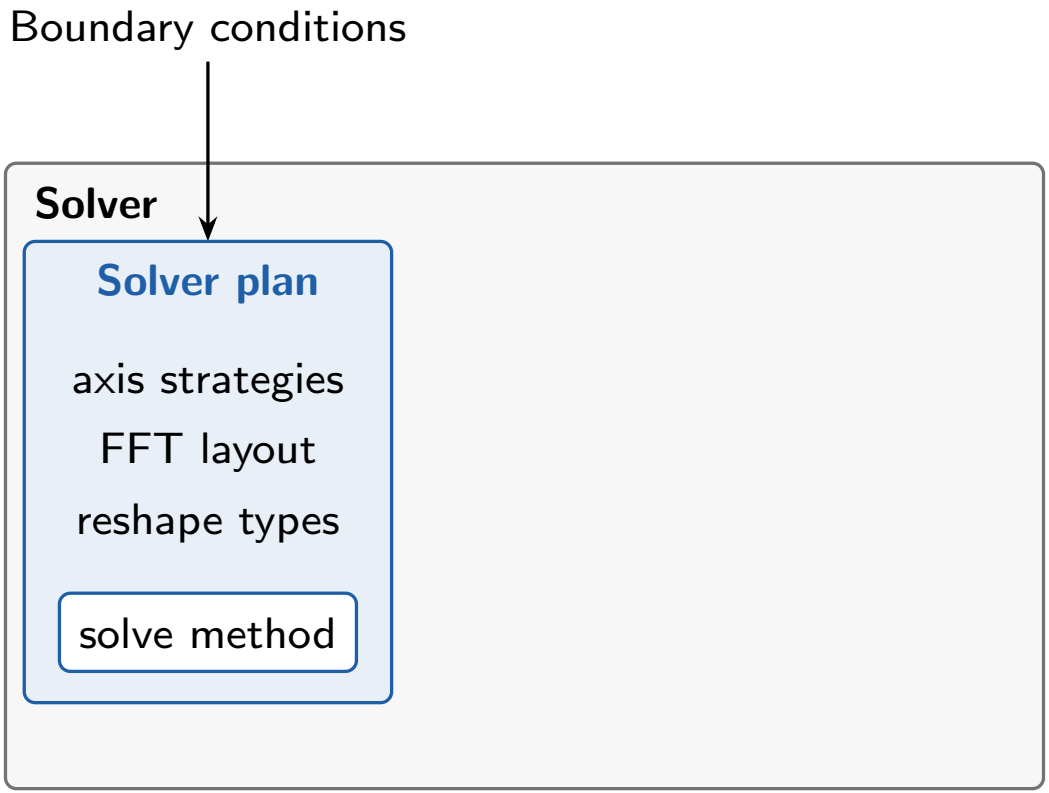
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3. Improve separation of concerns
4. Write documentation

The architecture of poisson-solver



- Configuration-based decisions
- Compile-time setup
- Implications of those decisions
- ⟶ setup
- - -➤ each solve

The architecture of poisson-solver



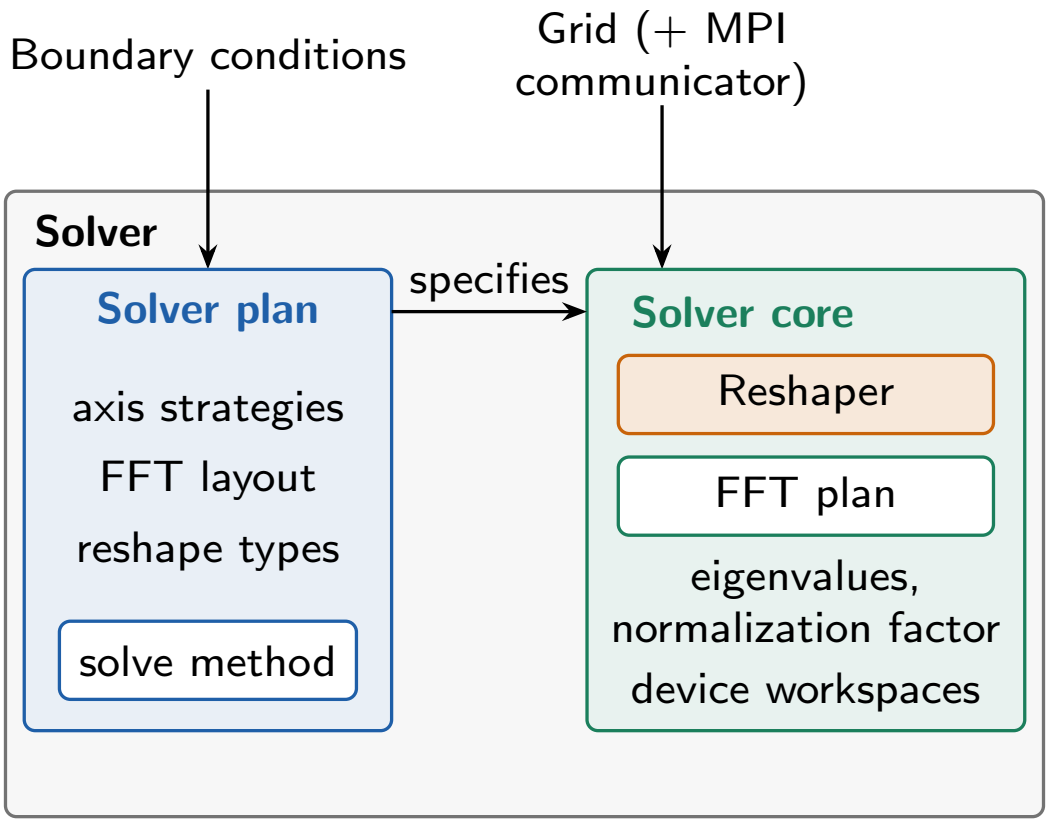
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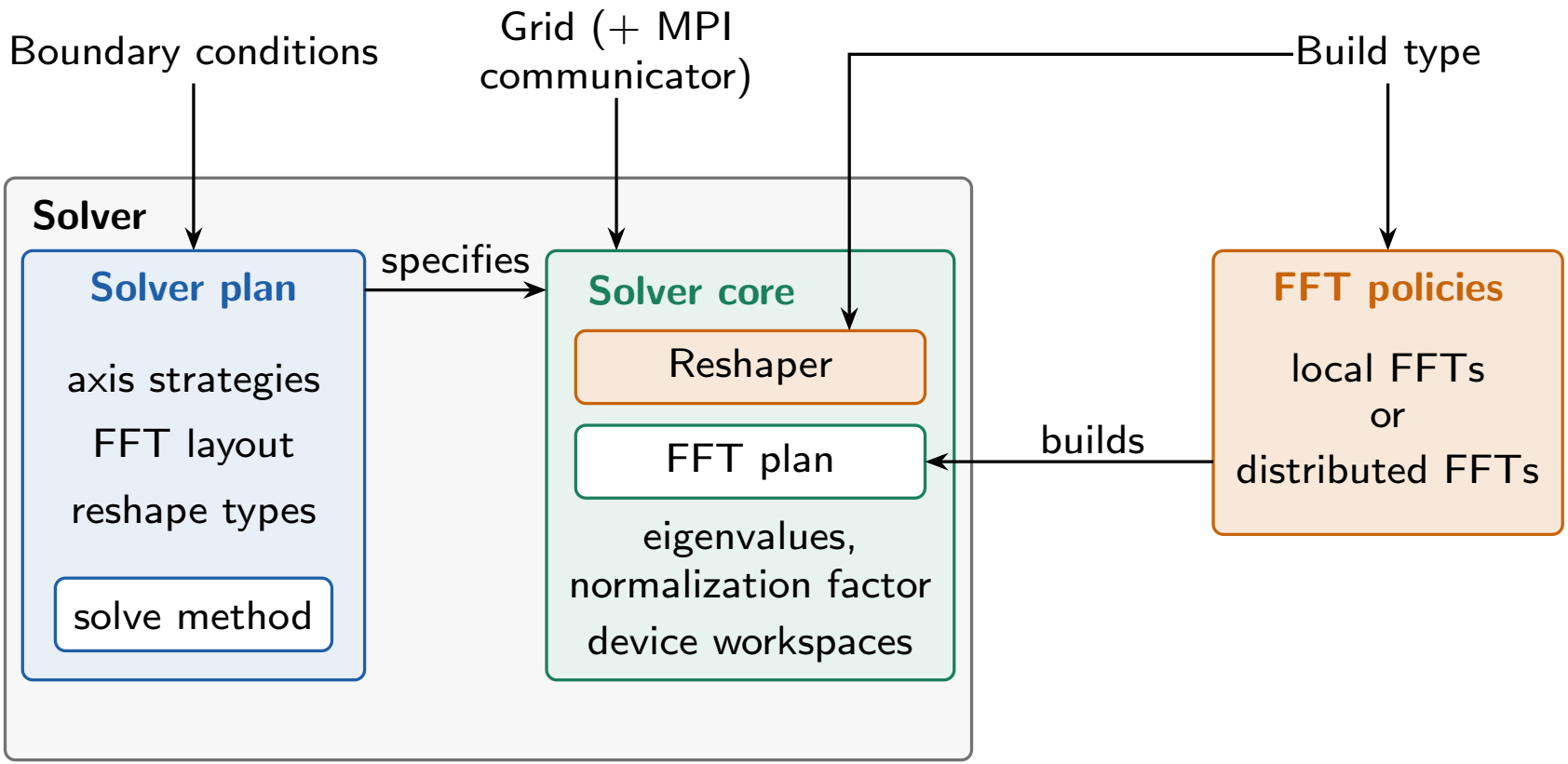
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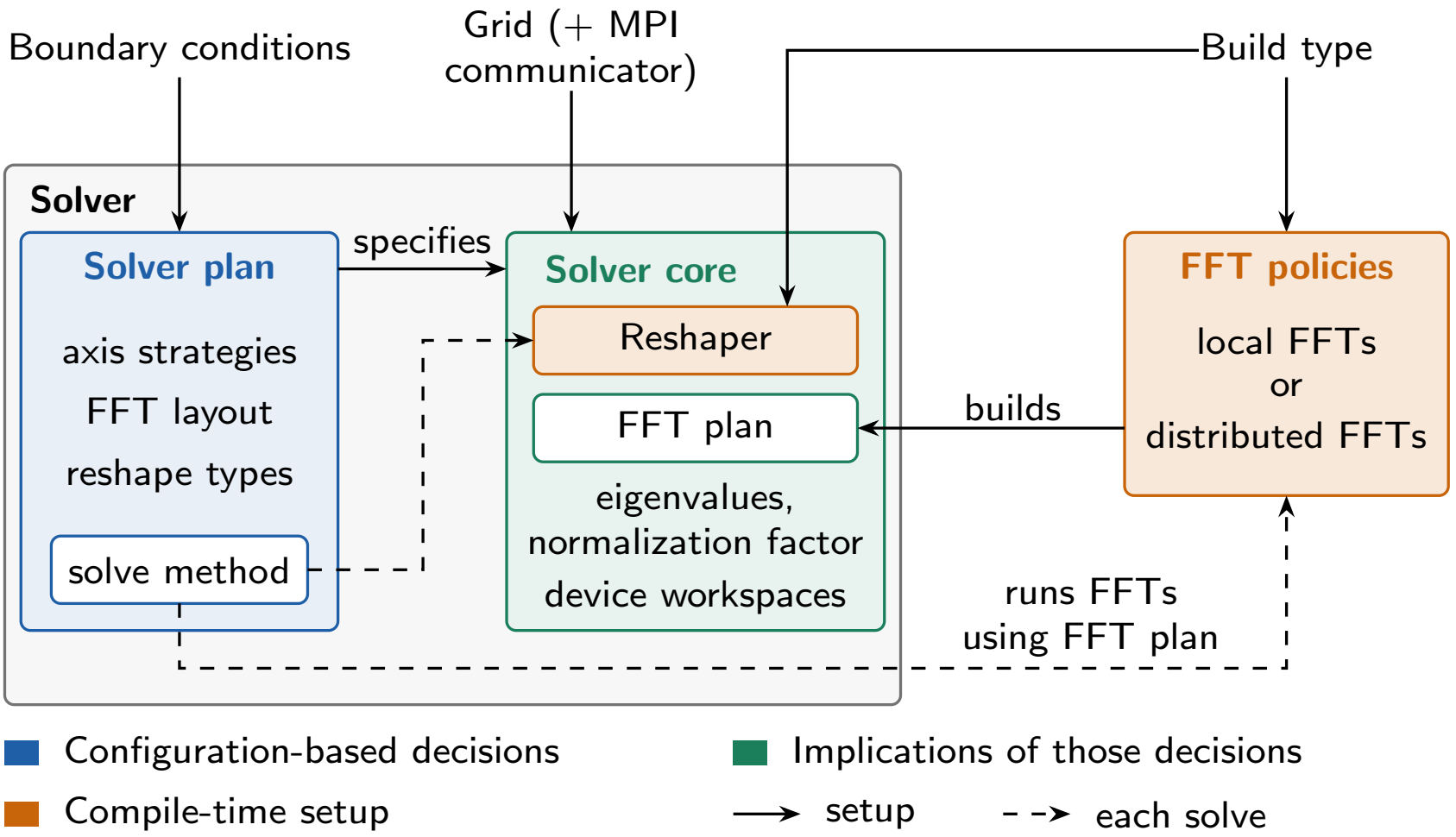


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- poisson-solver (device-device copies) always wins!
  - 2.4x to 10.4x speedups
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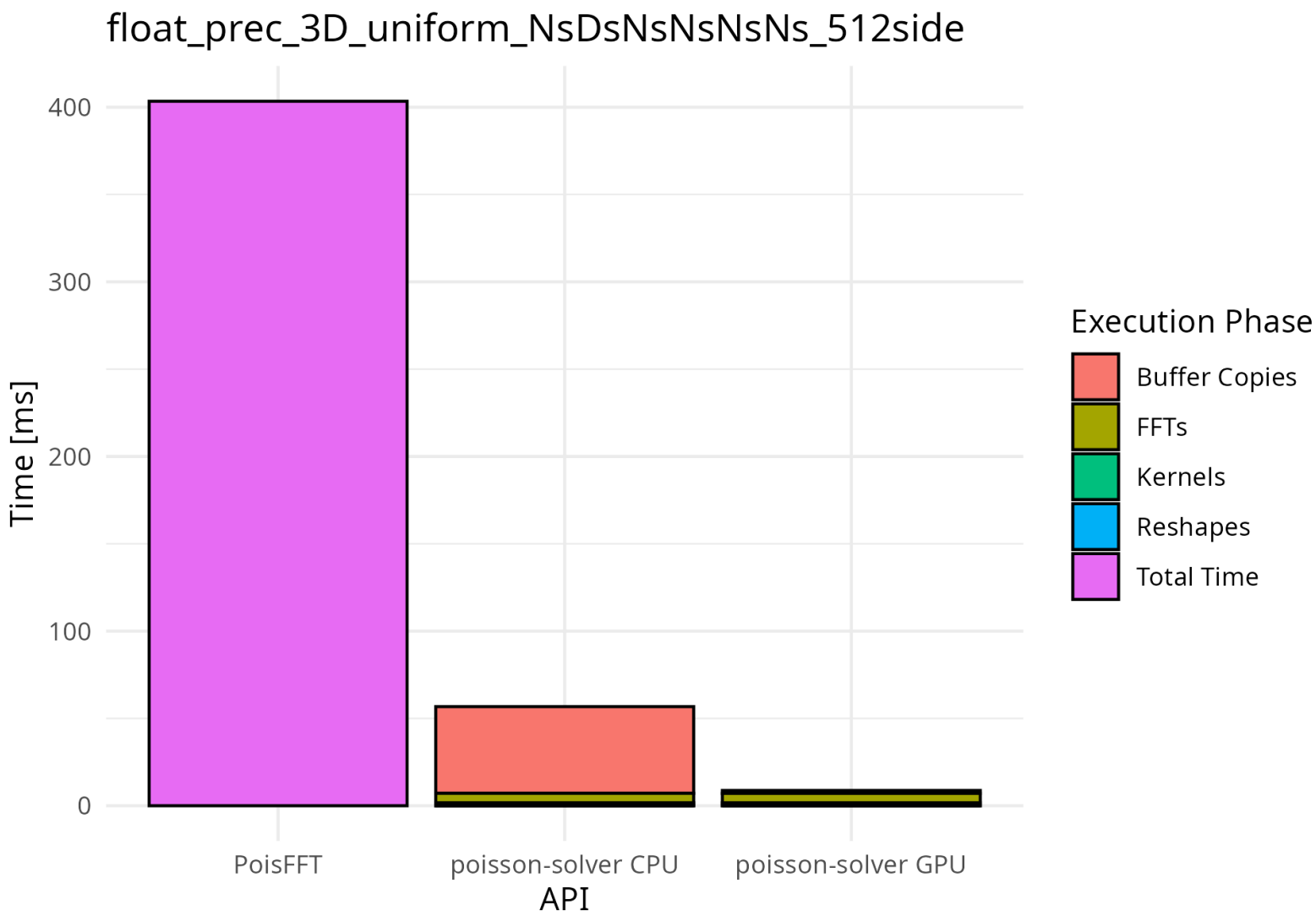
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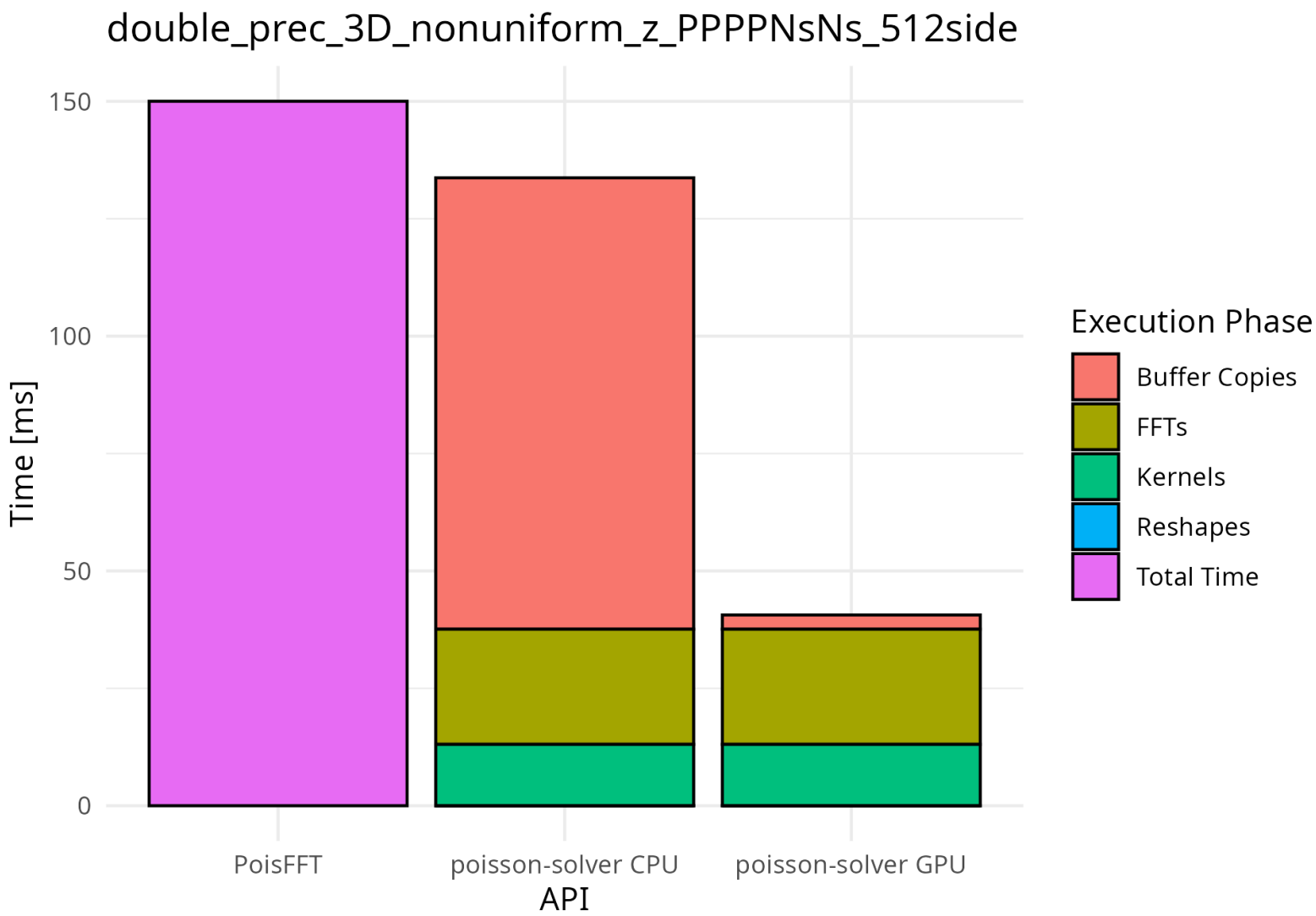
Distributed solver

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  - 1.31x to 2.36x speedups

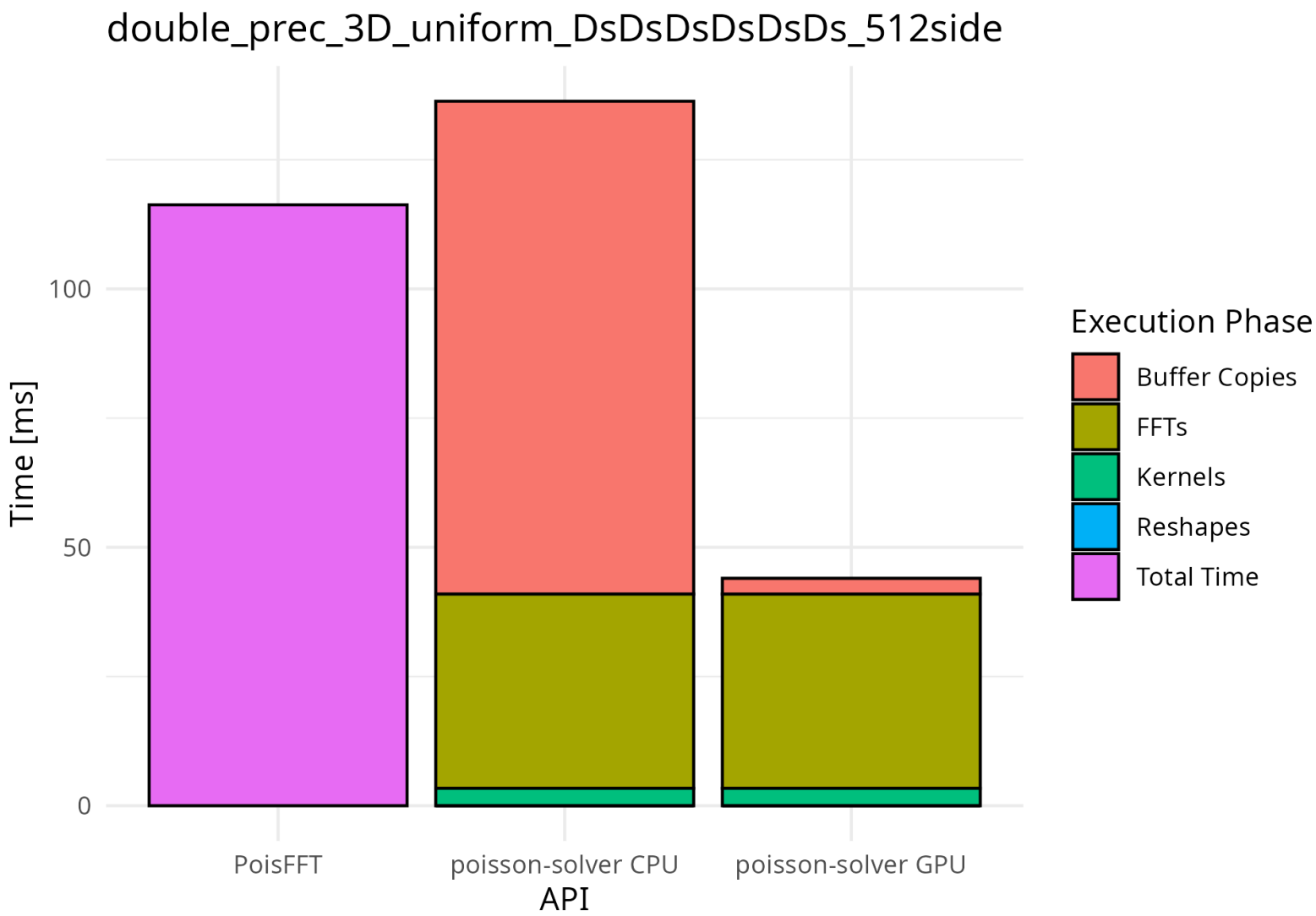
Local results



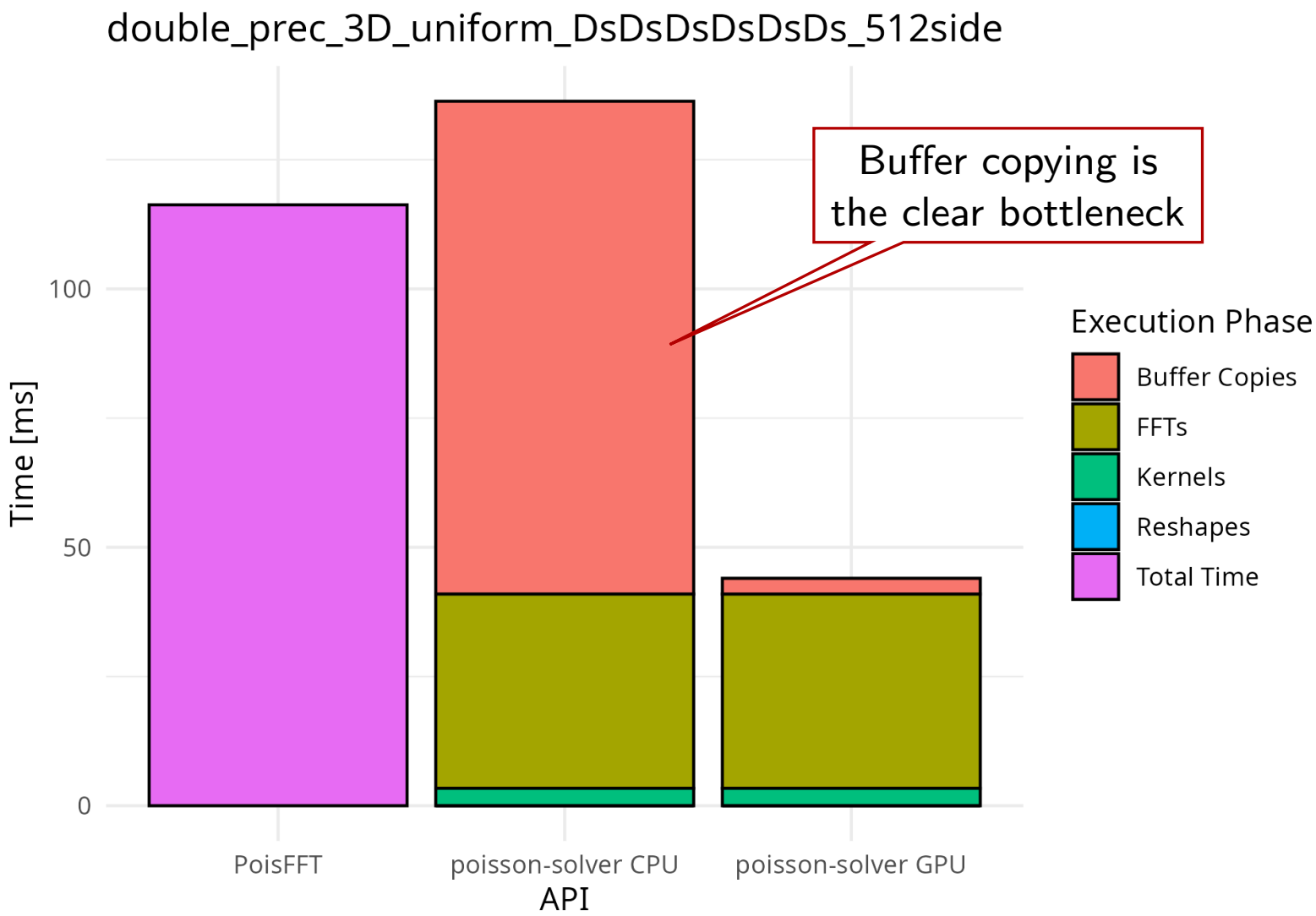
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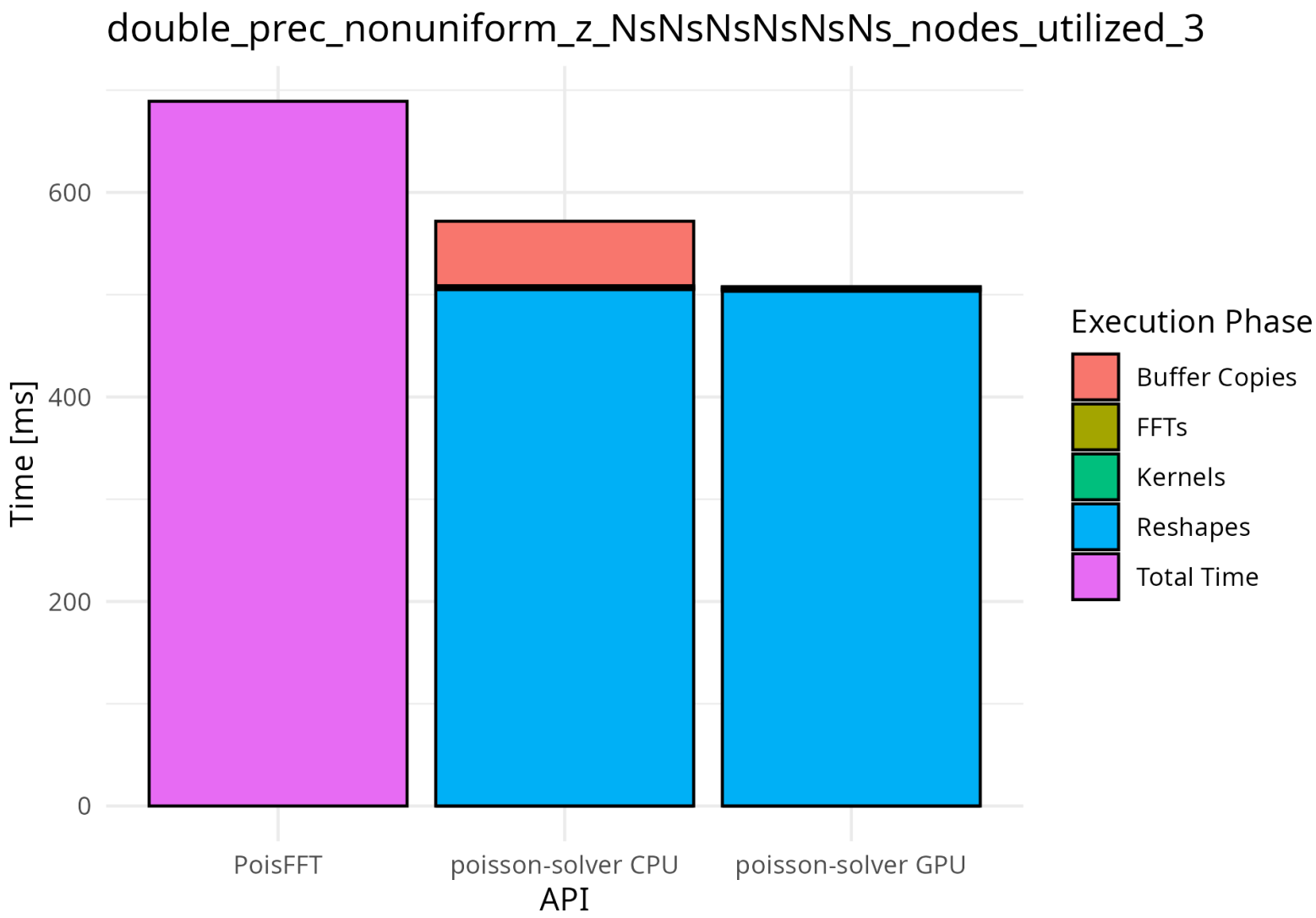
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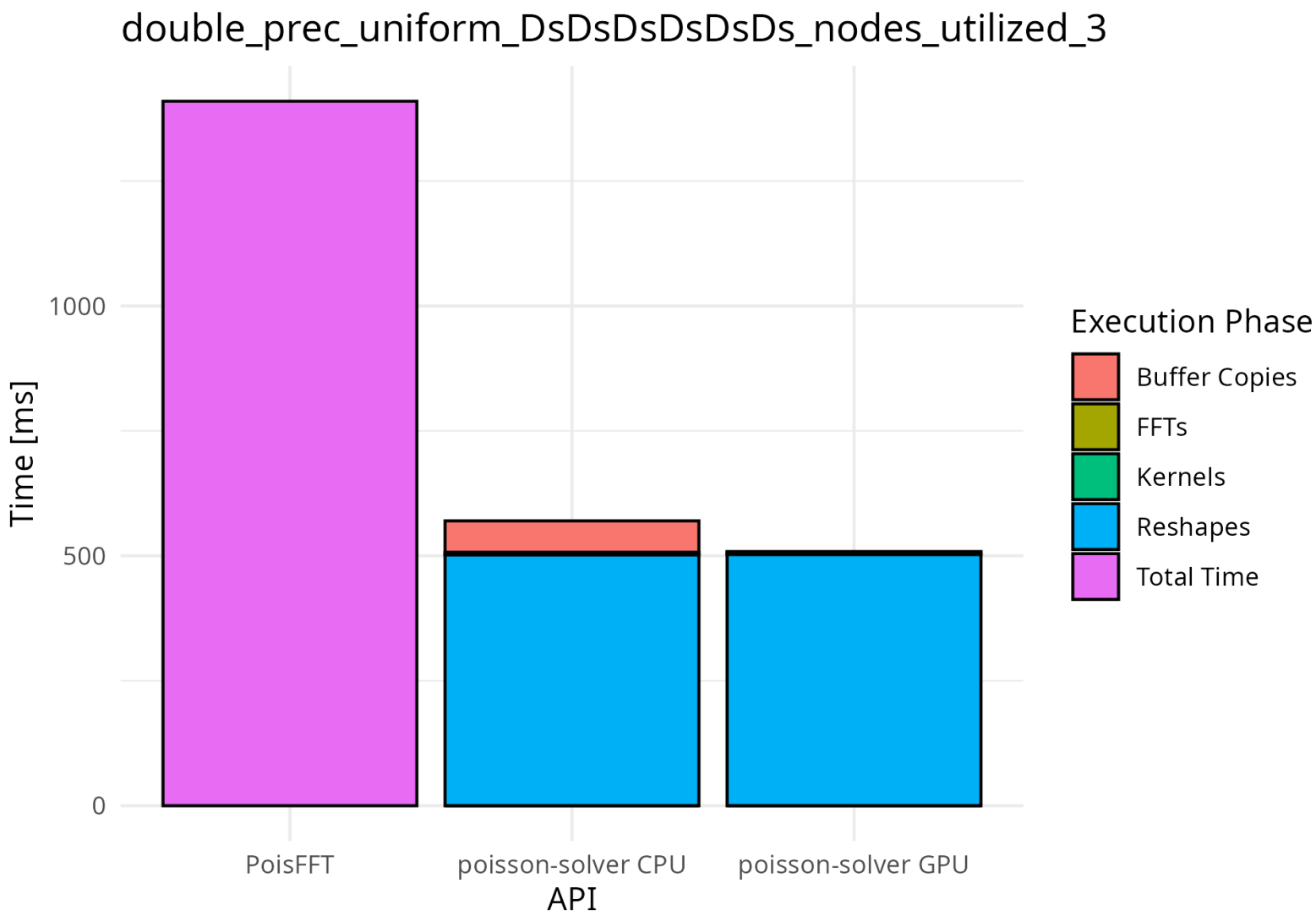
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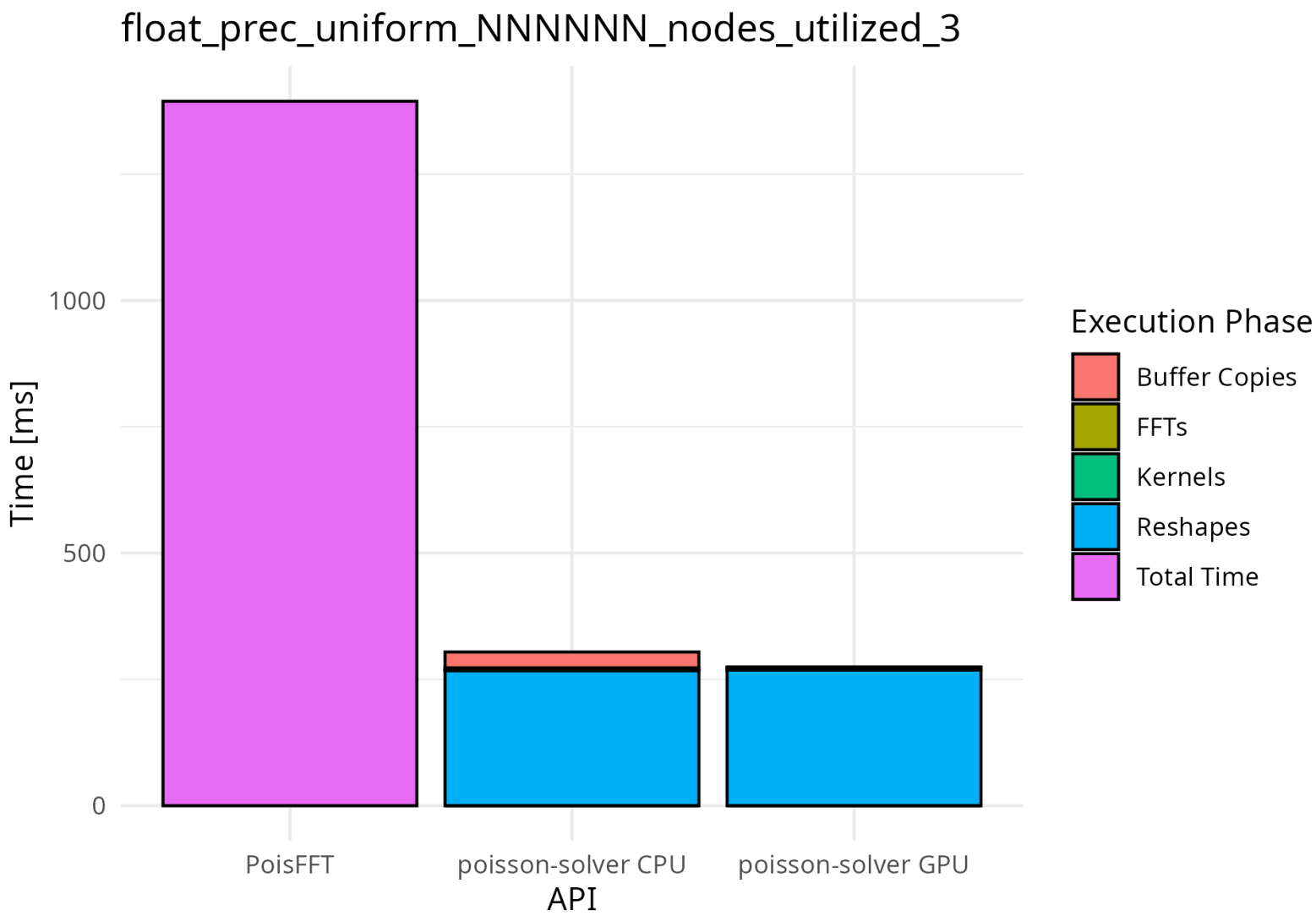
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Distributed results

- No clear bottleneck found
- But there is a problem in ELMM integration...

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Achievements and... what was hard?

- Ported and modernized a Poisson solver library
- Significantly improved software architecture
- Improved performance for:
  - All setups supplying GPU buffers
  - All distributed setups
  - Most local setups supplying CPU buffers
- Extended a distributed FFT library with new types of transforms

Thank you for your attention!

Computational fluid dynamics

Navier-Stokes equations

The Navier-Stokes equations are

$$\rho(\frac{\delta \mathbf{u}}{\delta t} + \mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu \nabla^2 \mathbf{u} + F$$

- fluid density ρ
- velocity vector field $\mathbf{u}$
- time t
- pressure field p
- dynamic viscosity μ
- external forces F

Poisson equation

The Poisson equation is

$$\Delta \phi = b$$

Future work

- Port more of ELMM onto a GPU

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- Don't do it by hand

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- Port more of ELMM onto a GPU
- Don't do it by hand
- Develop a system that compiles Fortran source code into C++/CUDA source code, handles buffer copying, etc.

GPU-based FFT libraries

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- CPU-based libraries offer a diverse support of DFTs
- GPU-based alternatives often lack such diverse support
- We found a single-node GPU FFT library
- We didn't found a distributed library with the required support...
- So we chose one and extended it!